314707003_2.22.R

CDH

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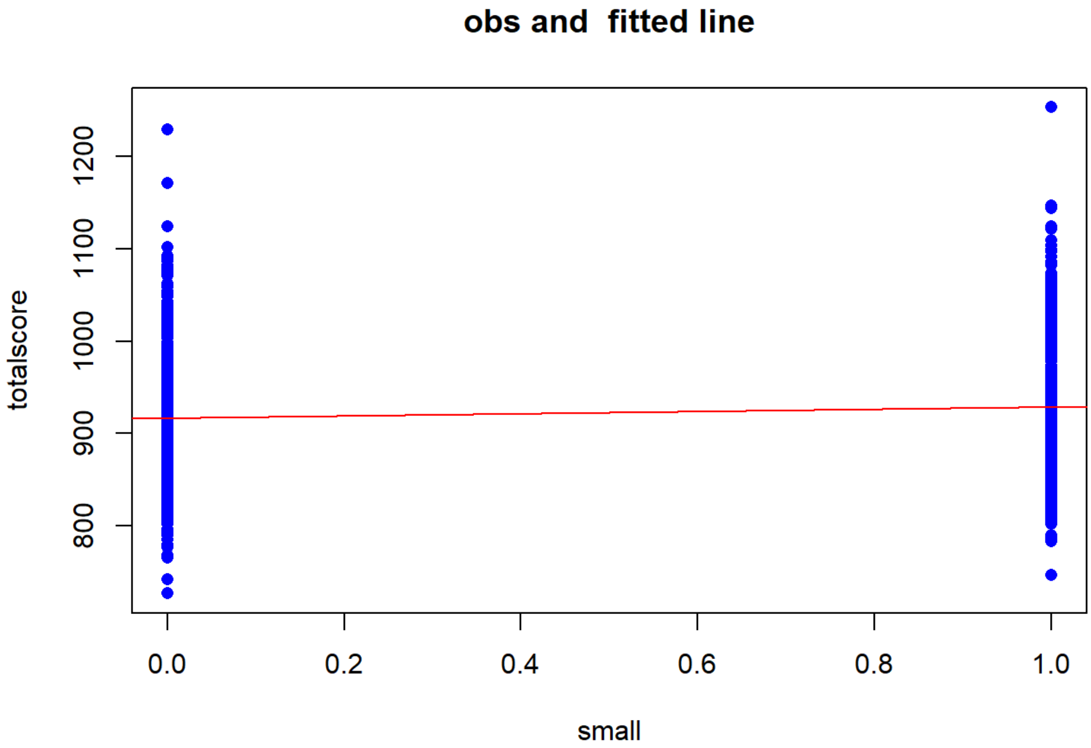
```
rm(list=ls())

#problem 2.22
url <- "http://www.principlesofeconometrics.com/poe5/data/rdata/star5_small.r
data"
dest <- file.path(tempdir(), "star5_small.rdata")

download.file(url, dest, mode = "wb")
load(dest)

#(a)
idx <- star5_small$regular == 1 | star5_small$small == 1


df <- star5_small[idx, ]
mod1 <- lm(totalscore ~ small, data = df)
intercept <- coef(mod1)[1]
slope     <- coef(mod1)[2]

###########################
plot(df$small, df$totalscore,
     xlab="small",
     ylab="totalscore",
     main = "obs and  fitted line",
     pch = 16,col = 'blue',
     type = "p")
abline(intercept,slope ,col = 'red')
```

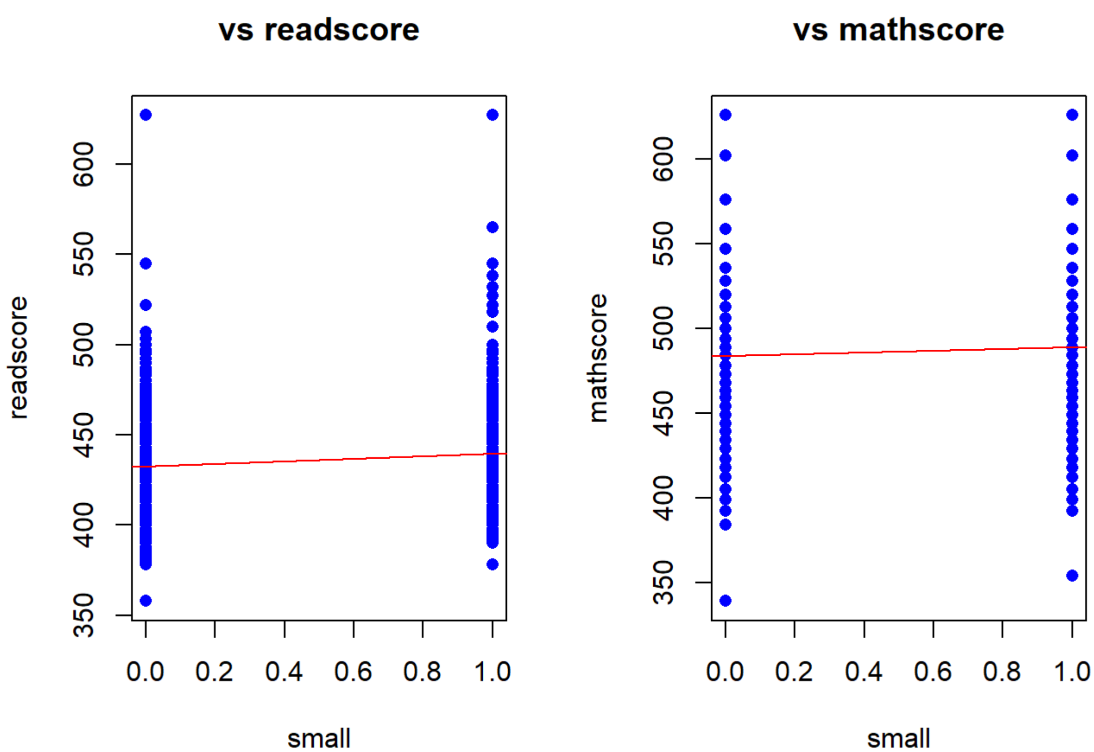



```
cat(
  "b1 is:", intercept,",when small=0,E[Totalscore|small=0]=b1", "\n",
  "b2 is:", slope,",when small increase 1(class is small),totalscore increase
b2", "\n",
  "since b2>0,small class has higher totalscore,but the points in the two gro
ups are widely dispersed.", "\n",
  "so it may not Statistically significant", "\n",
   sep = ""
)
```

```
## b1 is:916.4417,when small=0,E[Totalscore|small=0]=b1
## b2 is:12.17533,when small increase 1(class is small),totalscore increase b
2
## since b2>0,small class has higher totalscore,but the points in the two gro
ups are widely dispersed.
## so it may not Statistically significant
```

```
#(b)
mod2 <- lm(readscore ~ small, data = df)
intercept_2 <- coef(mod2)[1]
slope_2     <- coef(mod2)[2]


mod3 <- lm(mathscore ~ small, data = df)
intercept_3 <- coef(mod3)[1]
slope_3     <- coef(mod3)[2]
```

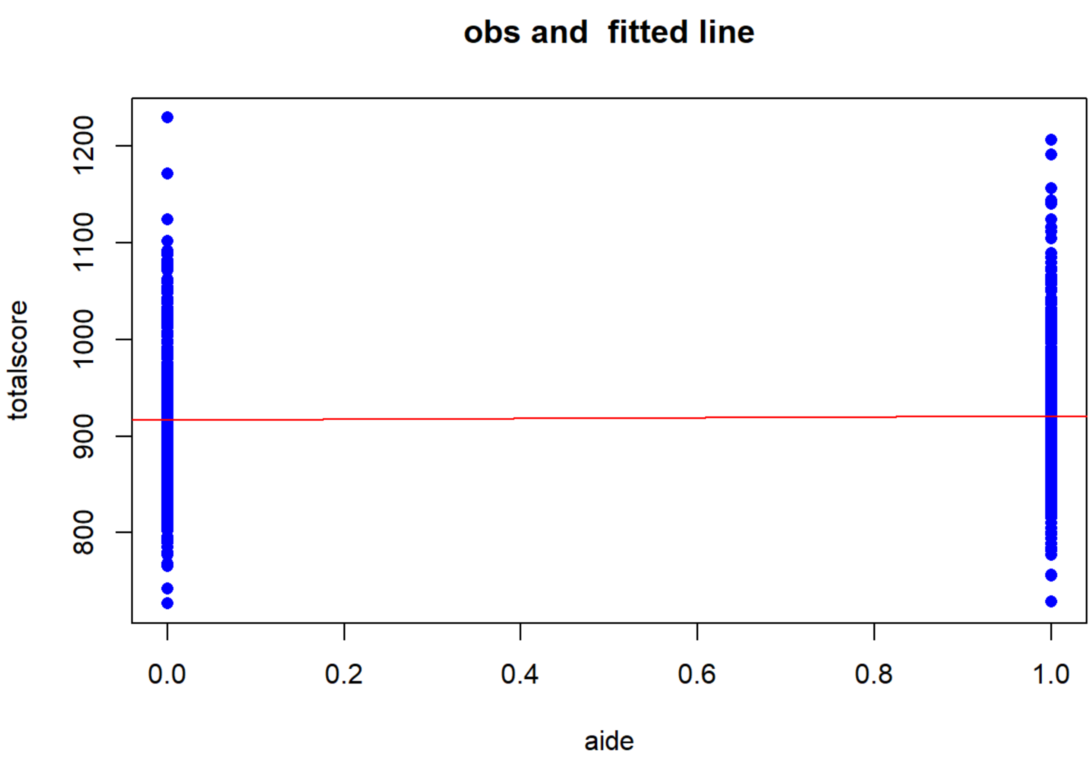
```
##########################
par(mfrow = c(1, 2))
plot(df$small, df$readscore,
     xlab="small",
     ylab="readscore",
     main = "vs readscore",
     pch = 16,col = 'blue',
     type = "p")
abline(intercept_2,slope_2 ,col = 'red')

plot(df$small, df$mathscore,
     xlab="small",
     ylab="mathscore",
     main = "vs mathscore",
     pch = 16,col = 'blue',
     type = "p")
abline(intercept_3,slope_3 ,col = 'red')
```




```
par(mfrow = c(1, 1))

cat(
  "b1_read is:", intercept_2,".b1_math is:", intercept_3, "\n",
  "b2_read is:", slope_2,".b2_math is:", slope_3, "\n",
  "readscore is more improved than mathscore when class is small", "\n",
  "but it may not Statistically significant", "\n",
  sep = ""
)
```

```
## b1_read is:432.665.b1_math is:483.7767
## b2_read is:6.924483.b2_math is:5.250849
## readscore is more improved than mathscore when class is small
## but it may not Statistically significant
```

```
#(c)
idx2 <- star5_small$regular == 1 | star5_small$aide == 1

df2 <- star5_small[idx2, ]
mod4 <- lm(totalscore ~ aide, data = df2)
intercept_4 <- coef(mod4)[1]
slope_4     <- coef(mod4)[2]

###########################
plot(df2$aide, df2$totalscore,
     xlab="aide",
     ylab="totalscore",
     main = "obs and  fitted line",
     pch = 16,col = 'blue',
     type = "p")
abline(intercept_4,slope_4 ,col = 'red')
```

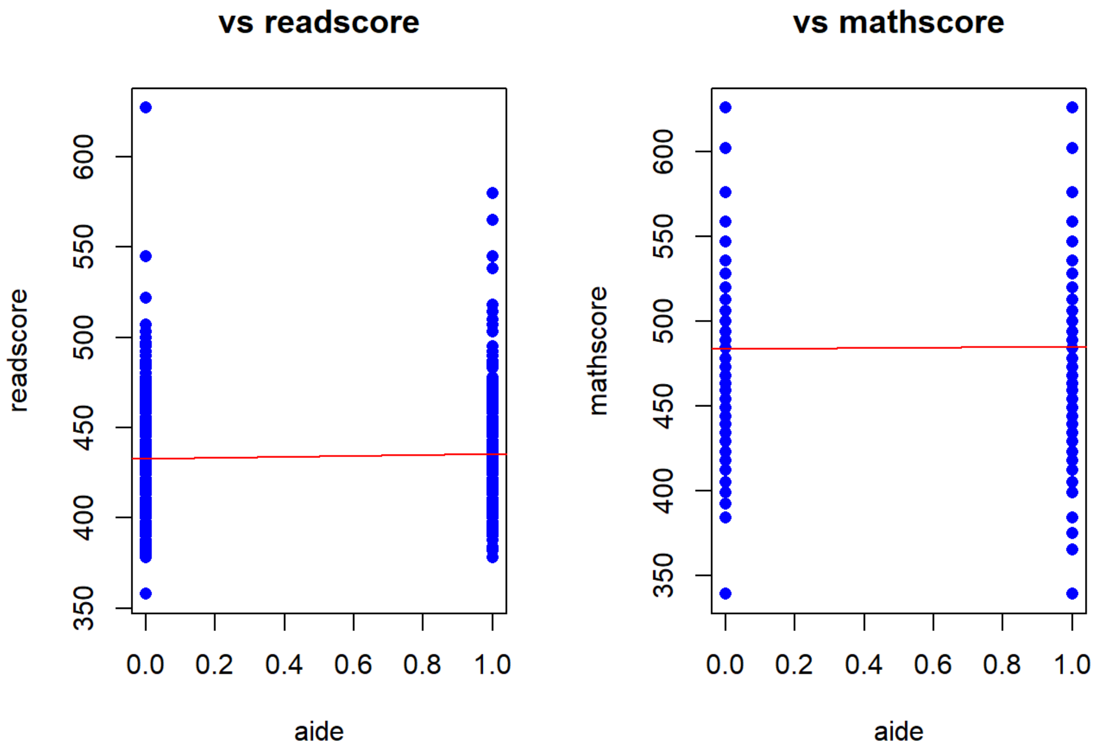



```
cat(
  "b1 is: ", intercept_4,",when aide=0,E[Totalscore|aide=0]=b1", "\n",
  "b2 is: ", slope_4,",when aide increase 1(class has aide),totalscore increa
se b2", "\n",
  "since b2>0,regular class with aide has higher totalscore,but the points in
```

```
the two groups are widely dispersed.", "\n",
  "so it may not Statistically significant", "\n",
  sep = ""
)
```

```
## b1 is: 916.4417,when aide=0,E[Totalscore|aide=0]=b1
## b2 is: 4.306488,when aide increase 1(class has aide),totalscore increase b
2
## since b2>0,regular class with aide has higher totalscore,but the points in
the two groups are widely dispersed.
## so it may not Statistically significant
```

```
#(d)
mod5 <- lm(readscore ~ aide, data = df2)
intercept_5 <- coef(mod5)[1]
slope_5     <- coef(mod5)[2]

mod6 <- lm(mathscore ~ aide, data = df2)
intercept_6 <- coef(mod6)[1]
slope_6     <- coef(mod6)[2]

###########################
par(mfrow = c(1, 2))
plot(df2$aide, df2$readscore,
     xlab="aide",
     ylab="readscore",
     main = "vs readscore",
     pch = 16,col = 'blue',
     type = "p")
abline(intercept_5,slope_5 ,col = 'red')

plot(df2$aide, df2$mathscore,
     xlab="aide",
     ylab="mathscore",
     main = "vs mathscore",
     pch = 16,col = 'blue',
     type = "p")
abline(intercept_6,slope_6 ,col = 'red')
```




```
par(mfrow = c(1, 1))

cat(
  "b1_read is:", intercept_5,".b1_math is:", intercept_6, "\n",
  "b2_read is:", slope_5,".b2_math is:", slope_6, "\n",
  "readscore is more improved than mathscore when class has aide", "\n",
  "but it may not Statistically significant", "\n",
  sep = ""
)
```

```
## b1_read is:432.665.b1_math is:483.7767
## b2_read is:2.871422.b2_math is:1.435066
## readscore is more improved than mathscore when class has aide
## but it may not Statistically significant
```